FE Graded Quiz M5

Time left: 59:07

QUESTIONS

1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16

Question 1

Why do we need to conduct convergence diagnostics for MCMC sampling?

- Convergence diagnostics are aimed at checking if outliers from MCMC samples are removed.
- Convergence diagnostics is aimed at checking whether MCMC samples are stationary.
- Convergence diagnostics is aimed at checking whether MCMC samples are normally distributed.
- Convergence diagnostics is aimed at checking whether MCMC samples are close enough to posterior distribution.

Question 2

We have the following variance equation for the GARCH(1,1) model:

$$\sigma_t^2 = 0.03 + 0.35r_{t-1}^2 + 0.45\sigma_{t-1}^2$$

Given $r_0 = 0.005, r_1 = 0.03$ and $\sigma_0 = 0.01$

Which of the following numbers is closest to the conditional variance for r at t=2?

- 0.066
- 0.022
- 0.033
- 0.044

Question 3

Which of the following measurement noise z_t is the right setup for standard normal GARCH(1,1) error for Kalman Filter?

- $z_t = s_t\epsilon_t$ where $\epsilon_t \sim student - t(0,1)$ and $s_t^2 = \alpha_0 + \alpha_1 z_{t-1}^2 + \beta_1 s_{t-1}^2$
- $z_t = s_t\epsilon_t$ where $\epsilon_t \sim N(0,1)$ and $s_t^2 = \alpha_0 + \alpha_1 z_{t-1}^2 + \beta_1 s_{t-1}^2$
- $z_t = s_t\epsilon_t$ where $\epsilon_t \sim N(0,1)$ and $s_t^2 = \alpha_0 + \alpha_1 z_{t-1}^2$
- $z_t = s_t\epsilon_t$ where $\epsilon_t \sim N(0,1)$ and $s_t = \alpha_0 + \alpha_1 z_{t-1} + \beta_1 s_{t-1}$

Question 4

Which of the following statements is **correct** regarding the ARCH(1) process?

- The ARCH(1) process is conditionally homoscedastic but unconditionally heteroscedastic when $0 \leq \alpha_1 < 1$.

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- The ARCH(1) process is conditionally heteroscedastic and unconditionally heteroscedastic when $0 \leq \alpha_1 < 1$.
- The ARCH(1) process is conditionally homoscedastic and unconditionally homoscedastic when $0 \leq \alpha_1 < 1$.
- The ARCH(1) process is conditionally heteroscedastic but unconditionally homoscedastic when $0 \leq \alpha_1 < 1$.

Question 5

The following table is a Bayesian update table for the foreign exchange rate GBP/USD and interest rate raise by the Bank of England.

hypothesis	prior	likelihood (raise interest | GBP/USD move)	Bayes Numerator	Posterior
Up	1/2	6/15		
Down	1/2	2/15		

"Prior" in the table means the probability of GBP/USD going up and the probability of GBP/USD going down. The likelihood column shows the probability of the Bank of England raising the interest rate given that the GBP/USD is going up and the probability of the Bank of England raising the interest rate given that the GBP/USD is going down. What is the correct (Up, Down) pair for Posterior?

- (6/15, 2/15)
- (6/30, 2/30)
- (1/2,1/2)
- (3/4, 1/4)

Question 6

Which of the following distributions is the most suitable error term distribution choice for asymmetric heavy tail asset return distribution to build a GARCH model?

- Normal distribution
- Binomial distribution
- Student's t-distribution
- Skewed Student's t-distribution

Question 7

Which of the following statements is **correct** regarding the states from Kalman Filter?

- The updated mean of the state is the simple average of the current mean of the state and the new observation.
- The updated mean of the state is the weighted average of the current mean of the state and the new observation.
- The updated mean of the state is the simple average of the current mean of the state and the gap of new observation vs. last observation.
- The updated mean of the state is the weighted average of the current mean of the state and the gap of new observation vs. last observation.

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Question 8

For an ARCH(1) process, which of the following ranges of α_1 will ensure that ARCH(1) has a fat tail distribution?

- $\alpha_1 > \frac{1}{3}$
- $0 < \alpha_1 < 1$
- $0 < \alpha_1 < \frac{1}{3}$
- $\alpha_1 > 1$

Question 9

Which of the following statements is **not** the main reason to use MCMC to sample posterior distribution?

- The posterior distribution is too complex to perform calculation involving integrals.
- There are several parameters in the model.
- The time series data have a fat tail distribution.
- The posterior distribution does not have a closed analytical form.

Question 10

Which of the following statements is correct regarding the adjusted Pearson goodness-of-fit test?

- This test is a normal distribution test.
- The test compares the empirical distribution of squared standard residuals with the selected theoretical distribution.
- The test only applies to the ARMA model.
- The null hypothesis (H_0) for the test is that the empirical distribution of the standard residuals is equal to theoretical distribution.

Question 11

We have the following state space model to model the movement of an asset return.

The state vector:

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

The initial values of states at t = 0 are assumed to be the following distributions:

$$x_1(0) \sim N(0, 0)$$

$$x_2(0) \sim N(3, 1)$$

The state equation:

$$\mathbf{x}_t = \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix} \mathbf{x}_{t-1} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mathbf{v}_{t-1}$$

Where $v_t \sim N(0, 1)$ and elements of v_t are mutually independent.

What are the means for x_1 and x_2 at t = 1?

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- (0.2,1)
- (0.6,3)
- (3,0.6)
- (1,0.2)

Question 12

For the following AR(1)-ARCH(1) model,

$$x_t = 0.3x_{t-1} + a_t\epsilon_t$$

$$a_t = \sqrt{0.9 + 0.2a_{t-1}^2}$$

Where $\epsilon_t \sim \textbf{normal iid white noise}(0,1)$, what is closest to the unconditional variance?

- 1.5
- 0
- 2.25
- 1.125

Question 13

When constructing a prior probability distribution, a researcher needs to do which of the following steps?

- Decide what posterior distribution would be best for manipulation and analysis.
- Select normal distribution for the prior.
- Collect all the knowledge and experience on the parameters and select an appropriate distribution for prior before studying the data.
- Review the data to be analyzed and select an appropriate distribution fit for the data.

Question 14

Which of the following statements is correct for GARCH model diagnostics?

- When comparing GARCH models with different specifications, we look for the model with the largest information criteria number.
- For Sign bias test, we look for a model with p value < 0.05.
- When comparing GARCH models with different specifications, we look for the model with the smallest likelihood number.
- For Nyblom stability test, we look for a model with a p value < 0.05.

Question 15

If we want to conduct pair trading for Apple stock and S&P 500 index, we need to find the hedge ratio of the two assets. Assume the true hedge ratio of the two assets is not observable. If we want to build a

state space model to describe the relationship of asset prices and hedge ratio, which of the following is the most accurate state space model?

- state equation: $\text{hedge ratio}_t = G_t * \text{hedge ratio}_{t-1} + w_t$; observation equation: $\text{Apple stock price}_t = \text{hedge ratio}_t * \text{S\&P 500 Index}_t + v_t$
- state equation: $\text{hedge ratio}_t = G_t * \text{hedge ratio}_{t-1} + w_t$; observation equation: $\text{Apple stock price}_t = F_t * \text{S\&P 500 Index}_t + v_t$
- state equation: $\text{hedge ratio}_t = \text{Apple stock price}_t * \text{hedge ratio}_{t-1} + w_t$; observation equation: $\text{Apple stock price}_t = F * \text{S\&P 500 Index}_t + v_t$
- state equation: ; observation equation: $\text{Apple stock price}_t = F_t * \text{S\&P 500 Index}_t + v_t$

Question 16

For the following AR(1)-ARCH(1) model,

$$x_t = 0.3x_{t-1} + a_t\epsilon_t$$

$$a_t = \sqrt{0.9 + 0.2a_{t-1}^2}$$

Where $\epsilon_t \sim \text{normal iid white enoise}(0, 1)$, which of the following values is closest to the $E(x_2|x_1 = 0.8, x_0 = 0.3)$?

- 0.3
- 0
- 0.8
- 0.24

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